



Semi-Autonomous CoreXY-Based Lug Nut Fastening/Unfastening System

Group S12
Capstone Project

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Abstract

This project presents the design and development of a semi-autonomous CoreXY-based system for lug nut fastening and unfastening. The objective was to improve consistency, efficiency, and safety compared to manual methods. The system integrates a computer vision module, CoreXY motion mechanism, and a drill-actuator subsystem to automate positioning and fastening operations. A Raspberry Pi and ESP32-CAM were used for bolt detection through image processing, while a microcontroller controlled synchronized stepper motor motion. Experimental testing demonstrated accurate bolt detection, reliable coordinate translation, and repeatable positioning of the drill mechanism. The system successfully performed fastening and unfastening operations with minimal human intervention, achieving stable motion at optimized speeds. While limitations such as calibration sensitivity and reduced performance in vertical orientation were observed, the prototype validated the feasibility of low-cost automation in automotive servicing. Overall, the system provides a scalable foundation for future fully autonomous tire-changing solutions.

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Table of Contents

<i>Abstract.....</i>	<i>2</i>
<i>Acknowledgements.....</i>	<i>2</i>
<i>Introduction.....</i>	<i>4</i>
<i>Literature Review / Background.....</i>	<i>4</i>
<i>Methodology / Design Approach</i>	<i>5</i>
<i>Implementation / Results</i>	<i>8</i>
<i>Discussion.....</i>	<i>10</i>
<i>Conclusion.....</i>	<i>11</i>
<i>Future Work / Recommendations</i>	<i>11</i>

Introduction

In automotive servicing and motorsport environments, the fastening and unfastening of lug nuts is a repetitive yet critical task typically performed manually using handheld tools. This process relies heavily on operator skill and consistency, often resulting in variations in torque application, reduced efficiency, and potential safety risks. With increasing demand for automation in mechanical systems, there is a growing need for solutions that can improve precision, repeatability, and overall workflow efficiency in real-world applications. Manual lug nut fastening presents several challenges, including inconsistent torque application, operator fatigue, and increased service time. These issues can lead to improper wheel fastening, which poses significant safety risks such as wheel loosening or component damage. Additionally, many existing automated solutions lack portability or are not suited for practical use in typical service environments. Therefore, there is a need for a compact, portable, and semi-automated system capable of accurately and consistently performing both fastening and unfastening operations while minimizing human error.

The objective of this project is to design and develop a semi-autonomous lug nut fastening and unfastening system that improves efficiency, accuracy, and reliability. The system aims to automate the positioning and fastening process using a CoreXY motion mechanism combined with computer vision and controlled motor actuation. Key goals include achieving repeatable and controlled torque application, reducing human error, enabling fast operation, and maintaining portability for real-world usage. This report is organized into several sections. The abstract provides a concise summary of the project. Section 1 introduces the background, problem, and objectives. Section 2 presents the literature review and relevant background. Section 3 outlines the methodology and design approach of the system. Section 4 details the implementation and experimental results. Section 5 discusses system performance, limitations, and key observations. Section 6 concludes the report, and Section 7 highlights potential future work and improvements.

Literature Review / Background

During the planning stages of project BoltBlitz, our team researched the existing solutions and technologies for tire changing systems. We came across automated and semi-automated systems in the market with a similar objective of changing tires, however, the key issues in each of these solutions are that they all only belong in industrial settings, they are not portable, and they are only made for professionals in automotive servicing. They also rely on large, expensive machinery integrating robotic arms and utilize pre-programmed motion to fasten and unfasten lug nuts. These solutions are effective but are inaccessible to most vehicle owners as they are not built for personal or portable use. Our system combines a CoreXY gantry system with a Computer Vision module for coordinate detection, and through our research, we noted a few hobbyist projects that use a similar integration of a CoreXY + Vision for different applications, however, our solution appears to be the only one targeted towards automotive servicing for lug nut fastening and unfastening. The current CoreXY + Vision integrated solutions mainly target 3D printing or robotic industrial applications, however, most existing solutions in the market incorporate just one of the two modules and rarely integrate the two to create a system that combines portability and affordability with vision-based detection.

The theoretical foundation involved in the design of BoltBlitz was based on several key electrical, computer, and mechanical engineering principles. The CoreXY physical hardware required a basic understanding of mechanical concepts including 3D printing components for structural design and building a belt and pulley system, whereas the control logic of the CoreXY required coordinating motor control to produce independent X-Y motion. It was also necessary to understand basic conversions between mm and motor steps in order to translate cartesian coordinates to and from motor step commands. The drill and actuator module also required similar foundational knowledge of motor and actuator control. The computer vision module incorporates key principles of image processing to detect lug nuts and translate all detected features into precise coordinates. From an electrical perspective there were many foundational concepts such as regulating current for the stepper motors, power conversion from AC to DC, and other critical electrical knowledge. Our project incorporated the theoretical foundation of multiple engineering streams through the integration of each of our modules, with mechanical design, motion control logic, and computer vision as a unified automated system.

As previously discussed, there are gaps in the automotive servicing industry, specifically referring to the tire changing process, which this project addresses. There is an availability of high-end and advanced tire changing machines, but there is a significant gap in the market for portable, cost-effective, and user-friendly solutions for quick roadside assistance or convenient at-home use. These systems are typically too large and complex for everyday users and vehicle owners, which severely limits the accessibility of these

systems. BoltBlitz addresses these gaps as it was designed and developed to be a compact, semi-autonomous system, integrating a Computer Vision module with a CoreXY gantry. Our project offers portability and scalability through a modular design where core modules are integrated yet independent. The future of this project would involve a feedback system and built-in databases for identifying precise torque for varying car models and therefore bridges the gap of current solutions with fixed configurations or pre-defined layouts. Finally, this project also addresses the gap of accessibility with current manual tire changes, reducing physical effort and technical expertise required for tire changes. Automating the process of fastening and unfastening lug nuts minimizes human error introduced with manual servicing, increases consistency, as well as making overall improvements to tire maintenance in safety, accessibility, and efficiency for a far wider range of users.

Methodology / Design Approach

Computer Vision + Detection Subsystem

The Computer Vision and Detection Subsystem was developed using integrated embedded hardware and computer vision software to detect the real-time real-world coordinates of the bolt centers for motor use and CoreXY alignment. This subsystem utilizes two hardware components including a Raspberry Pi 4 and an ESP32-CAM module, which were selected due to their coupled functionalities and costs. The ESP32-CAM was chosen as a suitable imaging device since it has an integrated camera module and a built-in web server, which was used for low-cost wireless image acquisition and transmission. The Raspberry Pi 4 serves as the main processing unit as it can support Python and OpenCV, which allows for the creation of computer vision and image processing algorithms. For this subsystem, the Raspberry Pi 4 generates its own Wi-Fi hotspot, allowing for communication between the Pi, ESP32-CAM, and laptop. The overall design approach includes a main script that follows a pipeline structured operation, which was organized into three sequential Python scripts: `system_check.py`, `camera_capture.py`, and `detection.py`. Prior to acquiring the image or performing the detection operation, the `system_check` script is run to verify that Raspberry Pi 4's Wi-Fi hotspot is running, the ESP32-CAM is connected to the hotspot, the ESP32-CAM's web server is responding, and the laptop being used for user interaction is also connected to the same network to allow for communication and full operation. The next script, the `camera_capture` script, handles capturing the image of the tire hub that will be utilized for the detection of the bolt coordinates. This script configures the camera settings including the image resolution (UXGA 1600x1200), JPEG quality, and onboard LED brightness level (150), which is done by sending HTTP control commands to the ESP32-CAM. Then, it provides the live stream URL of the web server and prompts the user to open it and position the tire hub in front. The script sends an HTTP request to obtain a still image from the camera upon user input and decodes it into an OpenCV image, which is saved on the Pi. The final script called `detection`, processes the OpenCV image and obtains the final bolt centroid coordinates. First, preprocessing including converting to grayscale and applying Gaussian blur (7x7 kernel) to reduce noise is applied. Binary inverted thresholding (value = 100) is used to isolate the dark regions in the image including the black bolt surfaces. Then, contour detection is applied to identify the bolt faces based on geometric constraints or filters including minimum area (80 px), maximum area (3000 px), and circularity threshold (≥ 0.6). Following this, the centroid coordinate (in pixels) of these detected bolt faces is calculated by using image moments, where the centroid x and y coordinates are determined by dividing the weighted sum of pixel positions in the horizontal direction and vertical direction by the contour area. The pixel coordinates are then converted into real-world millimeter coordinates using a scaling factor, where the pixels-per-millimeter ratio was the average pixel diameter divided by the bolt diameter (5 mm). The coordinates are then translated relative to the origin, which is the average center of all the bolts. Finally, the outputs are stored in a JSON file, a text file, and a debug image is produced. This debug image was used for

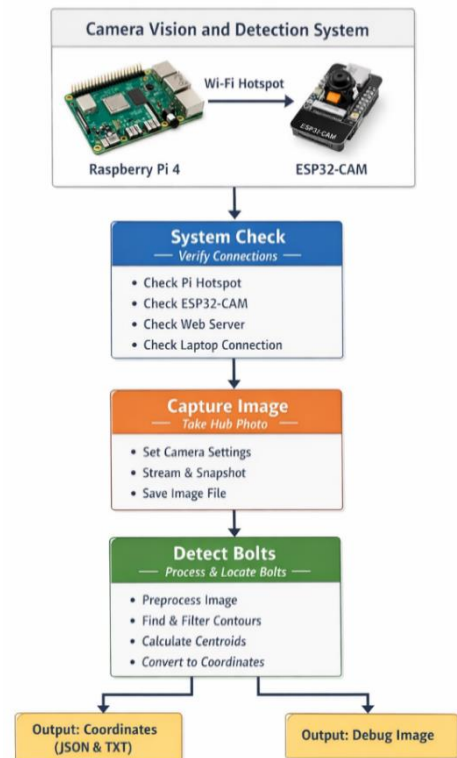


Figure 1. Flow of Computer Vision and Detection Subsystem

validation and data collection to see the precision and functionality of the detection script as it showed an overlay of the detected contours, bolt centers, and the computed origin. This image helped to determine whether certain parameters or thresholds needed to be adjusted. This approach was chosen as a result of its robustness and computational simplicity. The techniques applied including thresholding and contour detection were mostly reliable under different lighting conditions, and were easily modified if required. Rather than using machine learning or more complex methods, contour detection was able to use geometric properties to efficiently find the bolt coordinates in real-time.

CoreXY Subsystem

The CoreXY mechanical subsystem was designed to provide an efficient planar motion and precise positioning of the drill over lug nut locations. The system consists of NEMA 17 stepper motors driving GT2 timing belt pulleys (2mm pitch, 6mm width) with adjustable height on the motor shaft to align with the top or bottom belt. Linear motion is achieved using 250mm solid linear rods along the frame sides and 200mm hollow rods across the center to reduce the total weight of the carriage that is holding the drill. The rods are supported by LM8LUU linear bearings for low-friction translation, and lubrication is applied to the center rods to reduce scratching and friction. The mechanical structure was designed using Onshape to view the entire mechanism. Custom parts were 3D-printed at Thode Makerspace and fabricated from PLA, which can support threaded fasteners directly without heat-set inserts. Multiple design iterations and reprints were required to refine hole sizing and bearing fits. The belt system incorporates both toothed GT2 pulleys mounted on motor shafts and idler pulleys to guide the belt throughout the system. Belt retainers were used to clamp the belts securely to the carriage, ensuring a fixed connection between belt motion and carriage displacement. Maintaining adequate belt tension was critical due to the system's vertical orientation, as it helped counteract gravitational effects and prevent positional drift. Assembly was performed on a 1.5cm thick wooden base to provide structural support and portability to the system. To secure the components, we used fasteners, screws, and adhesives to prevent pulley slippage on shafts. Physical testing was conducted without simulation to validate the motion performance. Then the system was tested using programmed motion patterns and manual control inputs to evaluate positioning accuracy and repeatability. Several challenges were encountered during testing, including belt slipping, vibration of the system in certain directions, and misalignment of the frame. These issues were addressed through improved mechanical alignment by trial and error, increased belt tension, and optimization of operating speeds for each axis. Accuracy was ensured by maintaining a tight belt and precise structural alignment. The CoreXY architecture was selected over traditional gantry systems due to its reduced moving mass, higher speed, and improved efficiency, making it well-suited for a portable and fast-operating automated fastening system.

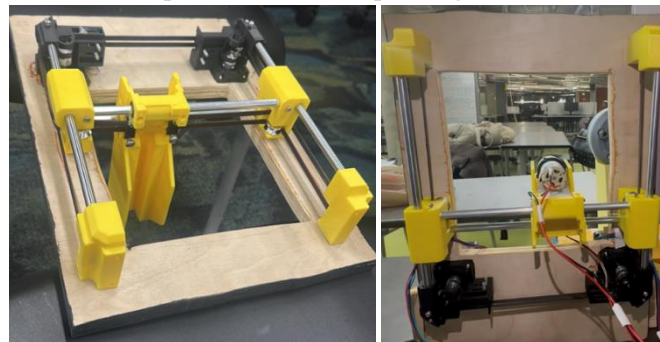


Figure 2. CoreXY Motion System (CAD model and Assembled Prototype)

CoreXY Control Logic Subsystem

The CoreXY motion and microcontroller sub-system was designed using a CoreXY gantry configuration enabling precise planar positioning using two NEMA-17 stepper motors. The CoreXY configuration requires both motors to work cohesively in contributing to the movement in both X and Y directions, rather than each motor controlling its own axis. This system is integrated into a belt-driven gantry along with motor drivers and a microcontroller responsible for the motion control logic of sub-system. The microcontroller was programmed to interface with higher level systems such as the computer vision sub-system, by accepting vision-based coordinate mapping text files as input and translating these coordinates into synchronized motor movement. This sub-system is not directly responsible for collecting data, however, it relies on coordinate inputs directly taken from the vision sub-system, which our team found was an efficient and modular alternative to combining both sets of code. This sub-system processes the coordinate data and converts them into motor commands, moving through each coordinate in a star formation as it is

done in typical manual tire changes. The hardware and equipment involved in this sub-system includes two NEMA-17 motors controlled by two Adafruit TMC2209 motor drivers that connect to the Raspberry Pi Pico 2 microcontroller. The software of this sub-system was developed in MicroPython for all motion control and logic, where all programming was done using Visual Studio Code with serial communication to the Pico 2. Additional hardware and tools used were an AC-to-DC power supply, breadboard wiring, and other cables. The approach that our team had taken for this sub-system was justified through the efficiencies and precision of the individual components and their ability to seamlessly integrate with each other. The stepper motors had enough torque for a proof of concept as well as smooth and precise control over motion and therefore made the approach straightforward without the need of complex feedback systems, especially paired with the motor drivers and microcontroller. This modularity of receiving an input text file and mapping between motor steps and metric measurements was convenient and therefore compatible for coordinate-based positioning processes. The general flow of the system architecture follows an efficient pipeline where it begins with receiving coordinate data through a text file sent by the vision system. The data is then processed and converted into relative or absolute displacements from the home position of the CoreXY gantry, and then the Pico 2 generates synchronized motor signals for both NEMA-17 stepper motors for fluid motion.

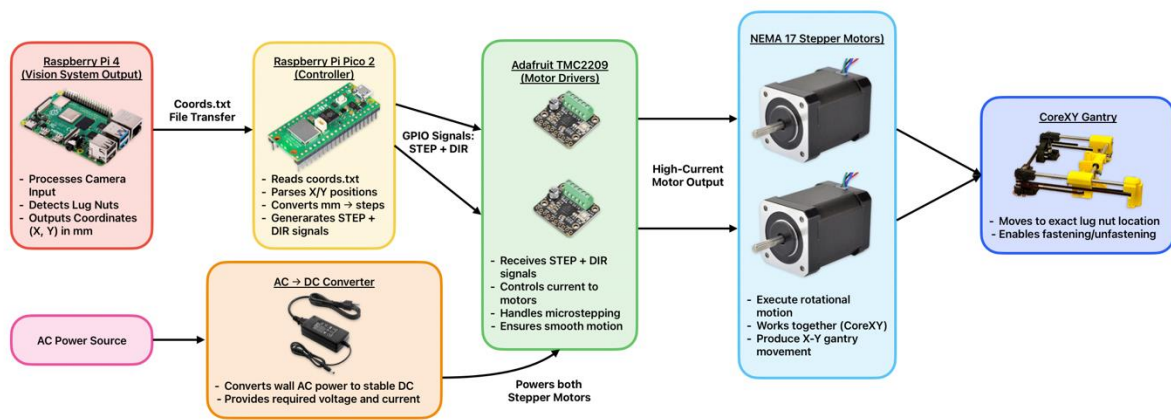


Figure 3. CoreXY Control Logic Module Flowchart

Drill Subsystem

The drill subsystem was designed using a modular approach to achieve controlled fastening and unfastening of lug nuts through coordinated rotational and linear motion. This subsystem integrates mechanical design, electrical control, and embedded programming to ensure precise engagement with the

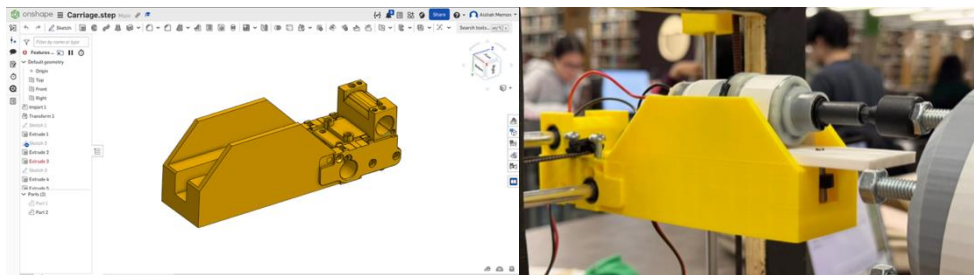


Figure 4. Final CAD Model and Physical Carriage Design

lug nuts. The mechanical design centers around a repurposed DC motor extracted from an IKEA power drill, selected for its high torque capability and compact form factor. The motor is mounted on a sliding plate that is connected to a linear actuator. A custom 3D-printed carriage was developed to securely house the actuator at the base of the subsystem. The sliding plate enables approximately one inch of forward and backward travel, allowing the drill to move into and out of contact with the lug nut. This motion is critical for maintaining proper alignment and preventing damage to the threads during operation. The electrical design consists of two dedicated motor drivers and a regulated power supply. A BTS7960 motor driver is used to control the DC drill motor due to its ability to handle higher current loads required for torque-intensive applications. A TB6612FNG motor driver is used to control the linear actuator, providing efficient bidirectional movement. Both drivers are powered by a 12V rechargeable LiPo battery, with a buck-boost converter stepping the voltage down to 6V to ensure safe and consistent operation across components. The

control system is managed by a Raspberry Pi Pico 2, which sends control signals to both motor drivers. This allows synchronization between the rotational motion of the drill and the linear positioning provided by the actuator. The actuator advances the drill to engage the lug nut and retracts slightly during fastening and unfastening to maintain alignment and reduce mechanical stress. Iterative testing and adjustments were conducted to optimize alignment, motion range, and response timing. Onshape was used for the CAD design of the carriage and structural components. PrusaSlicer and 3D printers were utilized for the fabrication and production of the parts. The Arduino IDE was used for initial coding, testing, and debugging of the control logic.

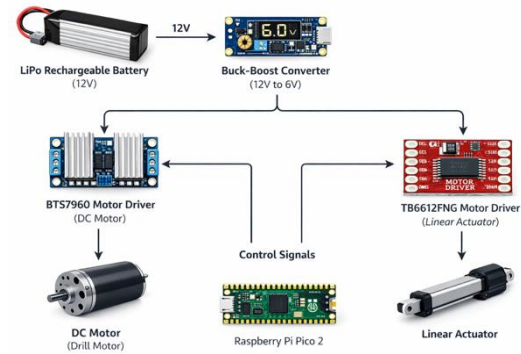


Figure 5. Drill System Architecture

Implementation / Results

Computer Vision + Detection Subsystem

The Computer Vision and detection subsystem was successfully implemented as a functional prototype using a Raspberry Pi 4, ESP32-CAM, and a laptop. For the physical setup of this subsystem, the ESP32-CAM was required to be at a fixed position facing the front of the tire hub to ensure consistency when capturing the image of the tire hub. To also ensure the tire hub was stable, it was mounted on a stand. An additional key part of the physical setup was painting the flat surface of the bolts black. By painting the bolts black, it became clearer for the camera to detect it in comparison to its original reflective metal. Furthermore, to increase the reliability of detection, we also used a blank white board and placed it behind the tire hub to eliminate extra background items or unwanted objects in the image. For testing, the first experiment was using varying light conditions. We observed that when the intensity of the ESP32-CAM's LED was high or at maximum brightness, the metal of the nuts or body of the bolts became highly reflective and bright and reduced accuracy, while low intensity caused missed detections. This is how the optimal brightness was determined to be 150 (range is 0 to 255). Another experiment done was testing the detection

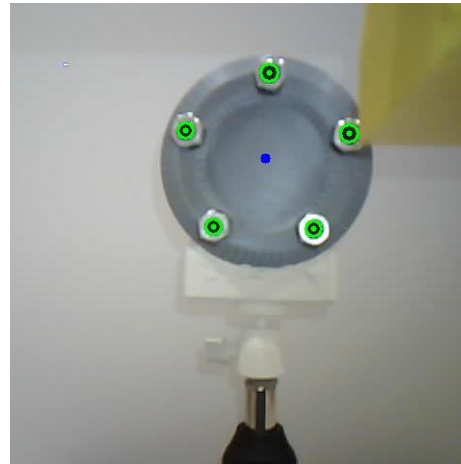


Figure 6. Tire Hub, Nuts, and Bolts Setup Figure 7. Outputted Debug Image of Bolts and Centroid of Bolts Detected

algorithm before and after painting the bolt surfaces black. Once there was more contrast for the algorithm to detect, the black faces of each bolt were detectable. The data analysis conducted through several trial-and-error attempts showed that there are a few main factors that impacted the detection accuracy. With lower and poorer image resolution, the shape of the bolt face became more obscured and difficult to detect. If the threshold value for detecting dark regions was too low or high, the algorithm would miss valid bolts or include unwanted noise, so we found 100 to be the best value. The circularity parameter (set to 0.6) is meant to help filter the contours detected so non-circular shapes are discarded. When the parameter was set to 1, some bolts were sometimes undetected as they don't appear as perfect circles in the image captured. For validation, we checked the debug image that was generated after execution of the detection algorithm. The accuracy of the coordinates generated was evaluated and tested by seeing if the motor guides the CoreXY to each bolt. We observed that while the detected coordinates were generally correct, minor adjustments needed to be made to the coordinates because of factors including possible shadows picked up, slight variation in

the carriage's home position, and differences between the angle of the tire hub in the image compared to its actual physical orientation. These factors created slight offsets and discrepancies between the actual and calculated coordinates. Overall, the complete Computer Vision and detection subsystem was able to operate end-to-end, beginning from capturing an image, processing it, and outputting the final millimeter coordinates.

Mechanical CoreXY Subsystem

The CoreXY system was fully assembled and integrated with the drill mounting, camera, and the tire hub. The final prototype included a rigid base, plexiglass shielding, and a hinged access door for safe operation and accessibility. The carriage successfully carried the drill and moved across the workspace using the completed belt and pulley system. Testing was performed with the drill attached to assess system behavior under load, using manual control inputs to evaluate system motion. Individual axis movements were tested first by commanding pure X and Y motion, followed by diagonal movement requiring synchronized motor operation. Repeatability was verified by repeatedly commanding the carriage to return to the same positions. The system achieved correct and coordinated motion in both X and Y directions. High positional accuracy was observed, with the carriage consistently reaching the intended lug nut positions. The system performed reliably within a speed range of approximately 6 mm/s. At higher speeds, particularly during diagonal motion, vibration and slight instability due to gravity were observed. These effects were reduced by lowering the operating speed. Tightening the belts to a taut condition eliminated slipping, prevented downward drift due to gravity, and ensured the belts remained properly engaged with the pulleys. After these corrections, the system demonstrated stable and repeatable motion.

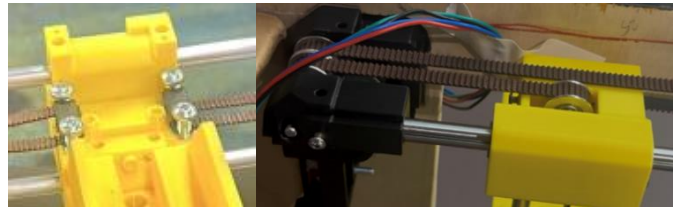


Figure 8. Belt Retainer Clamping Mechanism on Carriage and Timing Belt Routing through Idler Pulleys

CoreXY Control Logic Subsystem

The implementation of CoreXY motion and microcontroller subsystem involved many development stages that began with utilizing a basic stepper motor and motor driver from a previous course paired with the Pico 2 and Raspberry Pi 4. Basic testing with this setup allowed for a simple introduction to working with the Pico 2 and MicroPython before switching over to a dual-motor system with the NEMA-17 stepper motors and the TMC2209 motor drivers. In early development, it was assumed that each motor will control

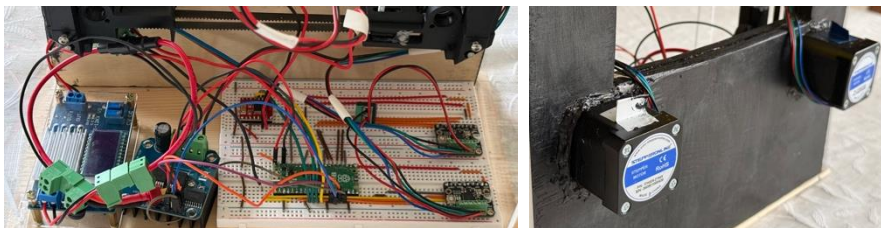


Figure 9. CoreXY Control Logic Breadboard Circuit and NEMA 17 Stepper Motors

one axis, however it was later understood that the CoreXY configuration requires both motors to work simultaneously for both X and Y directions, requiring revisions to the code. Software development also involved transitioning from basic steps/second to mm/second and added functionality of reading input coordinates. Experiments of the sub-system involved testing with pre-defined coordinates and verifying positional accuracy and the repeatability of the system. Testing was also done with the CoreXY gantry oriented vertically and horizontally to further evaluate system performance under different load distributions, where it was realized that vertical orientation had a large impact on the sub-system's ability to produce smooth and uninterrupted movement. Other experiments included manually pausing the sub-system to run the drill and motor's fastening/unfastening for simulation of an integrated system. As a result of these experiments, adjustments were made to the speed and acceleration of XY movement as well as motor control parameters to improve performance. Key data analysis performed was through testing, where system performance was evaluated based on specific criteria, including speed, accuracy, repeatability, and smoothness of motion. The vertical orientation of the CoreXY gantry was the significant limiting factor of

performance, as it impacted both speed and smoothness. However, after fine tuning system parameters and increasing tension in the pulley belts, we observed improvements in these areas.

Drill Subsystem

The drill subsystem was successfully developed as a functional prototype integrating mechanical, electrical, and control components. The system consisted of a 3D-printed carriage, a linear actuator, and a sliding plate mechanism supporting a repurposed DC motor from an IKEA power drill. All components were assembled and aligned to ensure smooth linear and rotational motion during operation. Testing was conducted to evaluate the performance of the subsystem in fastening and unfastening lug nuts. Experiments

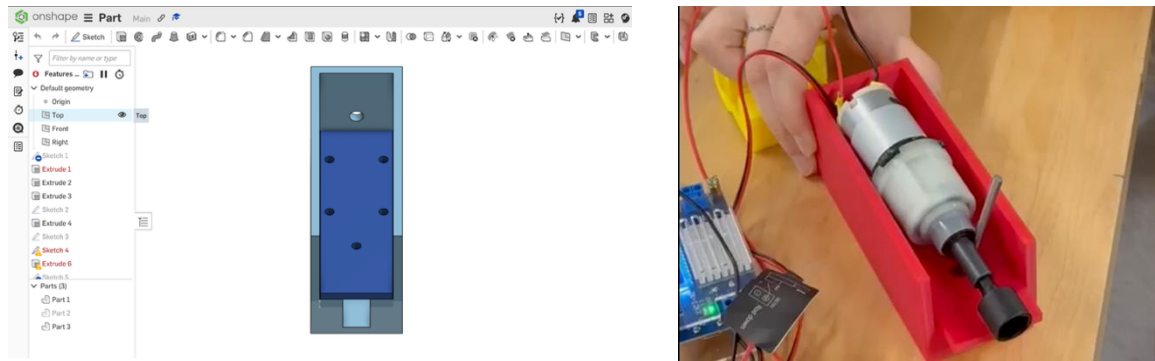


Figure 10. Initial Carriage Test

focused on verifying proper engagement between the drill and lug nuts, consistency of motion, and responsiveness of the control system. The linear actuator was tested to ensure accurate forward and backward positioning, while the DC motor was evaluated for sufficient torque and stability during operation. Iterative testing revealed that precise alignment between the motor and lug nuts was critical for reliable performance. Adjustments were made to the carriage design and actuator positioning to minimize misalignment and improve consistency. The coordinated control between the actuator and motor, managed by the Raspberry Pi Pico 2, resulted in smooth transitions between engagement, rotation, and disengagement. While no formal quantitative data analysis was required, qualitative observations indicated that the subsystem performed reliably under repeated testing conditions. The system consistently demonstrated the ability to engage, rotate, and disengage lug nuts with minimal manual intervention.

Discussion

The primary objective of this project was to design and develop an automated system capable of fastening and unfastening lug nuts with minimal human intervention. Based on testing and system performance, this objective was successfully achieved. The integration of mechanical, electrical, and control subsystems allowed for coordinated motion between the drill mechanism and positioning system, resulting in consistent and repeatable operation.

Compared to existing solutions, such as manual lug wrenches or commercial impact tools, this system introduces automation into a traditionally labor-intensive process. While industrial automated tire-changing systems exist, they are often expensive and complex. In contrast, this project demonstrates a cost-effective and scalable alternative using accessible components such as a Raspberry Pi Pico 2, repurposed motors, and 3D-printed structures. Although the system does not match the speed and durability of industrial solutions, it successfully demonstrates the feasibility of low-cost automation.

One of the key strengths of the system is its modular design. Dividing the system into mechanical, electrical, and control subsystems allowed for easier troubleshooting, iterative development, and future scalability. The use of a linear actuator to maintain alignment during fastening and unfastening improved reliability and reduced the risk of cross-threading. Additionally, the use of CAD software and 3D printing enabled rapid prototyping and design refinement.

However, several limitations were identified. A primary constraint was cost, which limited the selection of higher-performance components such as industrial-grade actuators, motors, and sensing systems. This resulted in trade-offs in speed, precision, and long-term durability.

Another significant challenge involved camera positioning and calibration. The system required a fixed and consistent distance between the camera, tire hub, and mechanical system in order to accurately translate visual coordinates into movement commands for the NEMA motors driving the CoreXY positioning system. Any variation in this distance introduced errors in positioning, making alignment with

the lug nuts less reliable. This highlighted the importance of precise calibration and rigid mounting in vision-based automation systems.

Additionally, gravity had a significant impact on system performance. Since the CoreXY system was mounted in an upright orientation, the moving components were subject to downward forces, causing slight sagging during operation. This introduced limitations on the maximum speed and acceleration the system could achieve, particularly when moving vertically or against the direction of gravity. As a result, motion speeds had to be reduced in certain directions to maintain accuracy and prevent misalignment. In contrast, if the system were oriented facing downward, gravity would assist the motion rather than oppose it, allowing for higher speeds and improved efficiency. This observation highlights the importance of system orientation and load distribution in the design of motion-based mechanical systems.

Several unexpected challenges were encountered throughout development. Achieving precise alignment required multiple iterations of both the mechanical structure and control logic. Synchronizing the motion of the CoreXY system, linear actuator, and drill motor also required careful tuning to ensure smooth and coordinated operation. Overall, the project successfully demonstrates the feasibility of an automated lug nut fastening and unfastening system. While there are limitations and areas for improvement, the system provides a strong foundation for future enhancements, including improved calibration methods, more rigid structural design, and higher-precision components.

Conclusion

BoltBlitz was successful in demonstrating the design and implementation of a semi-autonomous lug nut fastening and unfastening system through the integration of a vision module, a CoreXY gantry and motion control, a linear actuator, and an automated drill. Our system achieved precise lug nut detection, effective translation to real-world millimetre coordinates, reliable motion control, and a robust mechanical system that demonstrates a complete end-to-end workflow. Key insights we gained through this project were the importance of accurate calibration between each module as well as careful integration of all developed code. Examples of this include converting vision-based coordinates to real-world coordinates and translating these to physical motor movement while ensuring stable and accurate alignment. Integrating the code for computer vision, XY positioning, the drill, and linear actuator system, accentuated the significance of synchronization and how minor delays can impact the overall system performance. Finally, with the integration of the physical CoreXY gantry, it was clear that the vertical form of the system gave us little room for error in its structural integrity and XY motor control. Overall, BoltBlitz met all primary objectives in demonstrating itself as a functional prototype for a semi-autonomous and portable tire changing device, made for all users regardless of skill or experience. While minimal limitations were observed due to manual homing of the vertical XY plane, vertical load challenges, and other calibration inconsistencies, the system was able to perform reliably following iterative improvements. BoltBlitz reflects a successful integration of all system modules in both hardware and software, as well as demonstrating the result of strong system-level design and implementation.

Future Work / Recommendations

Our team, The Pit Crew, created a solid foundation with BoltBlitz that would allow us to further develop the system into a more robust and fully automated design. There are several software, hardware, and mechanical improvements that can be made to enhance the performance of BoltBlitz. Improvements include increasing belt tension to minimize vertical drag, using motors with higher torque or counterbalancing to reduce the effects of gravity, refining software synchronization between sub-systems, and perfecting triangulation in lug nut coordinate mapping. The next steps for developing BoltBlitz is to eliminate manual intervention between stages of the system by establishing real-time communication between the vision system and the motion controllers, and head into fully automated territory. Other next steps we considered was involving limit switches and memory capabilities to store home positions enabling our system to follow absolute positioning with certainty, without the errors of manual homing. Simpler immediate steps included creating a PCB for reliability and the compact form factor, as well as constructing casing and a base with polished materials for a more refined finalized look. For the long-term application of BoltBlitz, we believe our system has strong potential in real-world applications, specifically in automotive servicing. With further refinement, our system can simplify the tire changing process for both professionals and vehicle owners with its portable and autonomous design. The core concepts of BoltBlitz involves combining vision-guided motion coupled with automated fastening, which are concepts that can be applied to a wide range of maintenance operations or assembly lines. Our system demonstrates the broader impact and scalability that vision-guided systems offer beyond the initial scope that was intended.

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